Quiz 3 October 4th 2022 Chem 1127 Name (First, Last): ______________________

Points: ______/16 (17)

1. = Aktiv Chemistry Ch 3 # 77. Write the complete ground-state electron configuration (i.e. H: $1s^1$) for neon (Ne). (**1 pt**)

   1s2 2s2 2P6

2. = Aktiv Chemistry Ch 3 # 97. How many unpaired electrons are found in a ground-state atom of oxygen (O)? (**1 pt**)

   1s2 2s2 2P4   2P [↿⇂][↿][↿]

3. = Aktiv Chemistry Ch 3 # 105. Which element has the ground-state abbreviated electron configuration: $[Ar]4s^2 3d^{10} 4p^5$? (**1 pt**) Circle the answer below.
   a. As   b. Br   c. Se   d. Te   e. I

4. = Aktiv Chemistry Ch 3 # 105. Rank the following atoms in order of decreasing size (i.e., largest to smallest): Rb, F, Mg, B, N. (**1 pt**) Circle the answer below.

A) Rb > Mg > B > N > F   B) N > F > B > Mg > Rb   C) Rb > Mg > F > N > B

D) Mg > Rb > B > N > F   E) Rb > Mg > N > B > F

5. Two reactions are shown below, circle the one that corresponds to 1st ionization energy, or write **neither** if neither of them corresponds. (**1 pt**)

   a. $$X(g) \longrightarrow X^+(g) + e^-$$

   Ionization Energy

   b. $$X(g) + e^- \longrightarrow X^-(g)$$

6. Write the ions with the correct ionic charges that are observed for monatomic ions of Na, Al, and O. (**2 pts**)

   Na^+ O^{2-} Al^{3+}

7. = Aktiv Chemistry Ch 3 # 135. Which of the following series of isoelectronic ions (Mg^{2+}, N^{3-}, F^-, Si^{4+}) has the ionic radii in order of largest to smallest? (**1 pt**)

   A) $Mg^{2+} > N^{3-} > F^- > Si^{4+}$ B) $Mg^{2+} > Si^{4+} > F^- > N^{3-}$ C) $N^{3-} > F^- > Si^{4+} > Mg^{2+}$
   D) $N^{3-} > F^- > Mg^{2+} > Si^{4+}$ E) $F^- > N^{3-} > Si^{4+} > Mg^{2+}$

8. ~ Aktiv Chemistry Ch 3 # 142, 149. Predict & write the chemical formula for the three following ionic compounds: (**2 pts**)

   Cu^{2+} & P^{3-} Ag^+ & Cl^- Mg^{2+} & NO_3^-

   Cu_3P_2 $AgCl$ $Mg_3(NO_3)_2$

9. = Aktiv Chemistry Ch 4 # 35. Which of the following bonds is a polar covalent bond? (**1 pt**)

   A) Li – Cl   B) Cl – Cl   C) H – Cl   D) Li - Li

10. = Aktiv Chemistry Ch 4 # 32. Which of the following is an ionic compound? (**1 pt**)

   A) MgS   B) H_2S C) S_8 D) SO_2

11. = Chapter 4 Slide 11. Write the electron configuration for the C^{4-} anion? (**1 pt**) 10 e⁻s

$1s_2\ 2s_2\ 2P_6$

12. Name the following ionic compounds. (**3 pts**)

    a. NaCl

    Sodium Chloride

    b. FeO (Fe is iron)

    Iron oxide

    c. $CaSO_4$ (SO_4^{2-} is 'sulfate)

    Calcium Sulfate

13. We discussed this last Thursday, Bonus point! What is the ionic charge on Pb if it is in the ionic compound $PbCl_4$?

$Cl_4 = 4- \Rightarrow \boxed{Pb^{4+}}$